Dynamics of Orchestrator–Worker Large Language Model Systems

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Abstract

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Introduction

Multi-agent LLM frameworks have become a dominant paradigm for complex reasoning tasks, yet little is known about the macroscopic dynamics that emerge when agents communicate through a shared workspace. Prior work has characterised individual agent capabilities [TODO, 2024a], prompt engineering for multi-step reasoning [TODO, 2023], and benchmark performance of collaborative agents [TODO, 2024b], but the dynamical properties of the routing process itself have received less attention.

We approach this gap through the lens of switched dynamical systems [Liberzon, 2003]. Let $x(t) \in \mathbb{R}^d$ denote the workspace embedding at round t. The system evolves as

$$x(t+1) = f_{\sigma(t)}\big(x(t)\big), \qquad \sigma(t) \in \{R, C, S\}, \tag{1}$$

where $\sigma(t)$ is the routing (switching) signal and f_R, f_C, f_S are the Researcher, Critic, and Synthesizer operators respectively.

Contributions.

1. We formalise orchestrator–worker LLM systems as switched dynamical systems and identify the Synthesizer as a dissipative element.
2. We introduce a suite of routing-sequence metrics (block entropy, motif frequencies, inter-arrival times) and workspace divergence measures applicable to any multi-agent LLM system.
3. We show empirically that LLM orchestrators converge to low-entropy, Synthesizer-free limit cycles, independent of task.

Results

Experimental design

We implement a three-worker system (R, C, S) driven by four routing strategies and three Synthesizer placement schedules (Figure 1). Each experimental cell (task × placement × strategy × orchestrator temperature T_o) is replicated k times with deterministic seeds, yielding routing sequences $\boldsymbol{\sigma}^{(k)}$, workspace trajectories $\{x^{(k)}(t)\}_{t=0}^{T}$, and final answers.

Figure 1: **Experimental design.** Schematic of the orchestrator–worker architecture, Synthesizer placement conditions, and the metrics computed from each run.

Routing entropy

The order-k block entropy of the routing sequence is

$$H_k = -\sum_{w \in \{R,C,S\}^k} p(w) \log_2 p(w), \tag{2}$$

where $p(w)$ is the empirical frequency of block w pooled across replicates [Cover and Thomas, 2006].

Motif analysis

We pre-registered eight routing motifs prior to data collection to avoid post-hoc inflation of significance (Table 1).

Table 1: **Pre-registered routing motifs and observed frequencies.**

Motif	Interpretation	Mean count per replicate (orchestrator / random)
RC	Researcher–Critic alternation, period 2	
RCRC	Period-2 limit cycle	
RCSRCS	Three-phase rhythm (conjectured healthy)	
RSRS	Premature consolidation	
SS	Synthesizer doublet	
SSSS	Degenerate self-feeding	
RR	Researcher monoculture	
CC	Critic monoculture	

Workspace divergence

For K replicates sharing an experimental cell, the finite-time divergence-rate proxy at round t is

$$\Delta(t) = \frac{1}{\binom{K}{2}} \sum_{i<j} \big(1 - \cos(x_i(t), x_j(t))\big), \tag{3}$$

where $\cos(\cdot, \cdot)$ is cosine similarity in embedding space. Equation (3) is a stochastic analogue of the finite-time Lyapunov exponent [Strogatz, 2015]; increasing $\Delta(t)$ with t indicates sensitive dependence on the random seed.

Discussion

Methods

Agent architecture

Each worker receives a structured system prompt encoding its control-theoretic role (Supplementary Note 1) and operates on a shared plain-text workspace. The Researcher appends new content; the Critic appends adversarial commentary; the Synthesizer *replaces* the workspace with a compressed draft, implementing the update rule

$$x(t+1) = \begin{cases} x(t) \oplus \delta_w(t) & w \in \{R, C\} \\ \Pi\big(x(t)\big) & w = S, \end{cases} \tag{4}$$

where $\oplus$ denotes concatenation in text space and Π is the Synthesizer's projection.

LLM backend

All experiments use `claude-haiku-4-5-20251001` via the Anthropic Messages API with a three-attempt exponential-backoff retry. The model identifier is pinned to ensure reproducibility; silent upstream model updates are a known confound in closed-API studies [TODO, 2024c].

Embeddings

Workspace and final-answer embeddings are computed with `all-MiniLM-L6-v2` [Reimers and Gurevych, 2019] ($d = 384$, ℓ_2-normalised), loaded from a local checkpoint to eliminate API non-determinism.

Reproducibility

Seeds are derived deterministically via MD5 of the cell parameters, making every replicate exactly reproducible from the cell specification. Code and raw JSON results are available at `[repository URL]`.

Data availability

Raw trajectory JSON files and analysis scripts are deposited at `[repository URL]` and will be made publicly available upon acceptance.

Code availability

All experiment code is available at `[repository URL]` under the MIT licence.

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Author contributions

Competing interests

The authors declare no competing interests.

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